CURRICULUM VITAE

Vibhuti Gupta, PhD
Oct 1, 2026

CONTACT INFORMATION

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EDUCATION

2020 - 2021	Postdoctoral Fellow, University of Michigan, Ann Arbor, MI, USA
2015 – 2019	Ph.D. in Computer Science, Texas Tech University, Lubbock, TX, USA
2010 – 2012	M.Tech. in Computer Science, SRM University, Chennai, TN, India
2006 – 2010	B.Tech. in Computer Science, BIET, Jhansi, UP, India

ACADEMIC APPOINTMENTS

September 2025 - Present	Associate Professor (Tenure-Track) Department of Biostatistics & Data Science School of Public and Population Health University of Texas Medical Branch, Galveston, TX
July 2025 – September 2025	Associate Professor (Tenure-Track) Department of Computer Science & Data Science School of Applied Computational Sciences Meharry Medical College, Nashville, TN
January 2021 – June 2025	Assistant Professor (Tenure-Track) Department of Computer Science & Data Science School of Applied Computational Sciences Meharry Medical College, Nashville, TN
January 2021 – September 2025	Director mHealth Wearable Sensors Lab Department of Computer Science & Data Science School of Applied Computational Sciences Meharry Medical College, Nashville, TN
January 2020 – January 2021	Postdoctoral Fellow Department of Pediatrics Hematology/Oncology Division-Michigan Medicine University of Michigan, Ann Arbor, MI

August 2019 – December 2019	Instructor Department of Computer Science Texas Tech University Lubbock, TX
July 2015 – December 2019	Research Assistant Department of Computer Science Texas Tech University Lubbock, TX
July 2015 – December 2019	Research Assistant Institute for Measurement, Methodology, Analysis & Policy Texas Tech University Lubbock, TX

EMPLOYMENT (NON-ACADEMIC)

August 2012 – June 2015	System Engineer Tata Consultancy Services Noida, UP, India

HONORS AND AWARDS

2024	Researcher of the Year award, School of Applied Computational Sciences, Meharry Medical College to recognize exceptional contributions in research (Award and Meharry President recognition)
2017 – 2018	Graduate School Fellowship, Texas Tech University
2015 – 2019	Research assistantship, IMMAP, Texas Tech University
2015	Summer Research Departmental Scholarship, Texas Tech University
2012	Indian President award for securing First rank in Masters, SRM University

PROFESSIONAL WORK HISTORY AND TEACHING EXPERIENCE

TEACHING

- ❑ Teaching/Mentoring Responsibilities at UTMB
  - **Programming Foundations for Data Science (BIOS-6306)** – Fall 2026
  - **Special Topics (BIOS-6196):** Deep Learning -Summer 2026
  - Guest Lectures: Introduction to Visual Analytics in Healthcare (BIOS-6303), Visualization R Package: ggplot2, Department of Biostatistics and Data Science, Feb 5, 2026.

- ❑ Mentoring

  MPH Students:
  - Anette Garza : Capstone Advisor

  HPTM PhD Student:
  - Yoav Weber

  Postdoctoral Fellow
  - Weibin Zhang

  K23 Training Advisory
  - Gregory Brusola

- ❑ Teaching/Mentoring Responsibilities at Other Universities (Texas Tech University)

  - ❑ Mentoring

    MS Student
    - Sundaram Yadav (MS in Computer Science): Thesis Advisor

- ❑ Teaching/Mentoring Responsibilities at Other Universities (Meharry Medical College)

  - ❑ Teaching

School of Applied Computational Sciences (SACS):

- **MSDS 510**, Computer Programming Foundations for Data Science, Course Director, 3-credit hour, (Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022, Spring 2022, Fall 2021)
- **MSBD 540**, Introduction to AI in Healthcare, Course Director, 3-credit hour, (Summer 2025, Fall 2024)
- **MSBD 551**, Applied Machine Learning, Course Director, 3-credit hour, (Fall 2023)
- **MSDS 570**, Visualization and unstructured data analysis, Course Director, 3-credit hour, (Spring 2023, Summer 2023)
- **MSDS 565**, Predictive Modeling and analytics, Course Director, 3-credit hour, (Spring 2023, Fall 2022)
- **MSDS 535**, Further mainstream Programming languages for Data Science, Course Director, 3-credit hour, (Spring 2022)

School of Applied Computational Sciences (SACS) Curriculum Development:

- MSDS 510, Computer Programming Foundations for Data Science

    - Degree programs: MS and Ph.D. in Data Science, MS in Biomedical Data Science
  - MSDS 535, Further mainstream Programming languages for Data Science
    - Degree programs: MS in Data Science and Computer Science, Ph.D. in Data Science
  - MSBD 540, Introduction to AI in Healthcare
    - Degree programs: MS and Ph.D. in Biomedical Data Science
  - MSDS 545, Introduction to Computational Software Engineering
    - Degree programs: MS and Ph.D. in Biomedical Data Science
  - MSDS 555, Big Data Management & Analytics
    - Degree programs: MS and Ph.D. in Data Science, Ph.D. in Biomedical Data Science
  - MSBD 610, Network and Graph Theory for Biomedical Data Analytics
    - Degree programs: Ph.D. in Data Science and Biomedical Data Science
  - MSCS 540, Theoretical Foundations of Artificial Intelligence
    - Degree programs: MS in Computer Science
  - MSCS 565, Advanced Computer Systems
    - Degree programs: MS in Computer Science
  - MSCS 575, High Performance Computing
    - Degree programs: MS in Computer Science
  - MSCS 530, Principles of Programming Languages
    - Degree programs: MS in Computer Science

❑ Mentoring

Co-authors in peer reviewed journal/conference publications are shown in bold

Faculty/Peers:
- **Lubna Pinky** (Assistant Professor-Mentor, 2023-present)
- Mohammed Khan (Assistant Professor-Mentor, 2024-present)
- Asmah Muallem (Assistant Professor-Mentor, 2024-present)

Biomedical Data science Master's Students:
- Megan Leak (Spring 2025)
- Jean-Hus Theodore (Fall 2023 -Spring 2024), Project: Processing histopathology images for prostate cancer risk prediction, AIM-AHEAD grant
- Hannah Boykin (Fall 2023 -Spring 2024), Project: Exploratory data analysis on Fitbit data of Diabetes patients for All Of Us Data
- **Cyruss Tsurgeon** (Fall 2023-Spring 2024), Project: Exploring visualization approaches for RNA-seq data
- **Julian Broughton** (Fall 2023-Present), Project: Ethical and responsible AI in healthcare
- Destiny Pounds (Fall 2023-Present), Project: Ethical and responsible AI in healthcare
- Wajehah Sanders (Spring 2022), Project: Physionet challenge 2022

Data science Master's Students:

- **Farhana Manzoor** (Fall 2024-Present), Project: Exploring visualization approaches for RNA-seq data
- Xiangye Xiao (Fall 2023 -Spring 2024), Project: Exploring radiology images for prostate cancer risk prediction
- **Ange Rukundo** (Fall 2023-Present), Project: Ethical and responsible AI in healthcare
- **Lorrayya Williams** (Fall 2023-Present), Project: Detecting virus integration sites in tumor genomes using deep learning
- **Aleesa Mann** (Fall 2022 – Fall 2024), Project: Unmasking COVID through vocal biomarkers
- Shara Tylor (Spring 2022), Project: Physionet challenge 2022

Biomedical Data science and Data Science PhD Students:

- Courtney Quarterman (Spring 2025 - Present), Project: Designing the k-12 training programs
- Ernest M. Taku (Spring 2025 - Present), Project: Optimizing Diabetes Management with Personalized Insulin Dosing Using Reinforcement Learning
- Firoj H. Shaikh (Spring 2025 - Present), Project: Multimodal approaches for Prostate cancer risk prediction
- **Emirrah Sanders** (Fall 2021, Fall 2024 -Present), Project: Improving Detection of Virus Integration in Tumor Genomes Using Deep Learning

Capstone Committees:

- **Julian Broughton** (Summer 2025), Project: Learning Unbiased Risk Prediction Based Algorithms in Healthcare
- **Destiny Pounds** (Summer 2025), Project: Forecasting Minute-by-Minute Stress, Anxiety, and Affective States Using Time-Series Analysis of Wearable Sensor Data
- **Lorrayya Williams** (Summer 2025), Project: A Novel Pipeline for Virus Integration Sites Detection in Tumor Genomes Using Deep Learning
- **Aleesa Mann** (Summer 2023), Project: Unmasking COVID through vocal biomarkers
- **Ange Rukundo** (Summer 2025), Project: Diesel Prices Trends Analysis and Forecasting with Machine Learning on State and National Levels in U.S. (2020 – 2024)

High School Student:

- **Shreyansh Baidya** (Fall 2024): Worked on project *"*Predicting Cardiac Complications of Myocardial Infarction Patients Using Machine Learning*"* and published in IEEE Big Data Conference 2024

❑ Teaching/Mentoring Responsibilities at Other Universities (University of Michigan and Texas Tech University)

Mentoring

Postdoctoral Fellows:

- Rajnish Kumar, University of Michigan, Ann Arbor (Fall 2023 – Present),

Project: Developing AI/ML ready data for mHealth and wearables
- Bengie L. Ortiz, University of Michigan, Ann Arbor (Fall 2023 – Present), Project: Developing AI/ML ready data for mHealth and wearables

Capstone Committees:
- Baby Aneesha, Texas Tech University (Fall 2021), Project: Hybrid CNN-LSTM for MRI Brain Tumor Image Classification
- Vamsidhar Reddy, Texas Tech University (Fall 2021), Project: Classification using deep learning neural networks for brain tumors
- Satraboina Alekhya, Texas Tech University (Fall 2021), Project: Convolutional Neural Networks for skin cancer detection using Transfer learning

❑ Program Development of Educational Meeting or Workshop at Other Universities (Meharry Medical College)

International/National
- Session on Generative AI for efficiency in Higher education and institutional research, SACSCOC Annual Meeting, December 2024, Austin, Texas, Role: session design and speaker
- NSF Expand AI leadership Workshop (EXAIL) in Summit for AI Institutes Leadership (SAIL), October 2024, Pittsburgh, Pennsylvania, Role: planning committee member and speaker.

State/Regional
- Trustworthy AI tutorial for medical students Seminar, August 2024, Role: Organizer and speaker.
- NASA Collaborative interactive Data Science Academy for k-12 students, July 15-19, 2024, Role: planning committee member and speaker.
- NIH R25 Summer Data Science Bootcamp, June 2023 and June 2024, Role: planning committee member and speaker
- Ethical and Responsible AI in Medical Systems Seminar Series for medical and data science students, February – April 2024 and February – April 2025, Role: Organizer and speaker.
- Trustworthy AI in medical Systems Planning Workshop, March 2024, Role: Organizer and speaker.
- NASA Meharry Data Science Academy, July 17-19 2022, Role: planning committee member and speaker

❑ Development of Educational Enduring Materials
- Trustworthy AI teaching modules for K-12 educators' course (9 modules course), 2024, Links:
  - Webpage link: https://sacsmeharry.org/taims/#tab-education
  - GitHub repository link: https://github.com/vibgupta123/Trustworthy-AI-teaching-modules-for-High-school-educators
- Ethical and responsible AI webinar series, 2024, Links:
  - Recordings and material: https://sacsmeharry.org/taims/#tab-seminars

    - GitHub repository link: https://github.com/vibgupta123/Ethical-and-Responsible-AI-seminar-series
  - Trustworthy AI in medical systems planning workshop, 2024, Links:
    - Recordings and material: https://sacsmeharry.org/taims/#tab-workshops
  - Written article published in CXOTech magazine on the role of mHealth and Wearables in Disease Diagnosis in Healthcare, 2024, Link:
    - https://cxotechmagazine.com/role-of-mhealth-and-wearables-in-disease-diagnosis-in-healthcare/
  - NIH R25 Summer Data Science Bootcamp, 2023, Link:
    - Recordings and material: https://sites.google.com/view/nih-r25-talk/home
    - GitHub repository link: https://github.com/vibgupta123/NIH-R25-Talks

❑ Invited Educational Seminars/Lectures

International/National Meetings

1. Title: "*A Novel Efficient and Scalable Approach to Process Large-Scale Histopathology Data for Prostate Cancer*", AIM-AHEAD Southeast Hub Research Symposium, Nashville, TN, Mar 26-27, 2026.

2. Title: "*A Novel Generalizable Approach to Integrate Multimodal Data for Prostate Cancer*", AIM-AHEAD Southeast Hub Research Symposium, Nashville, TN, Mar 26-27, 2026.

3. Title: "*Extracting Actionable insights from Multimodal Healthcare data*", Workshop on Agentic AI for Healthcare: Ethics, Accountability, Resilience, and Trust (Agentic-HEART), IEEE EMBS BHI 2025, Atlanta, GA, Oct 26-29, 2025.

4. Title: "*A Novel Pipeline using UNI and CLAM for prostate cancer risk assessment for Histopathology*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, July 08-09, 2025.

5. Title: "*Automated histopathological pipeline for prediction of prostate cancer risk using nuclei-based WSI tiling*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, July 08-09, 2025.

6. Title: "*Integrative Machine Learning Approaches for Predicting Prostate Cancer Risk Using Multi-Omics Data*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, July 08-09, 2025.

7. Title: "*Modeling Burn Subtypes Using Human Centered AI*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, July 08-09, 2025.

8. Title: "*Predictive Modeling of Non-Medical Determinants of Health in Cancer*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, July 08-09, 2025.

9. Title: "*Panel: AI, Data and Cyberinfrastructure for Community Resiliency*", MS-CC Planetary Resilience Workshop: Tools, Training, and Collaboration Workshop, Jun 11-12, 2025.

10. Title: "*Ethical Considerations for AI in Healthcare Research*", Virtual NIH Data Science Connect Workshop, Jan 13-15, 2025.

11. Title: "*Predicting Cardiac Complications of Myocardial Infarction Patients Using Machine Learning*". The REU Symposium, In 2024 IEEE International Conference on Big Data (Big Data) Dec 15-18, 2024.

12. Title: *"A Novel Pipeline for Virus Integration Sites Detection in Tumor Genomes Using Deep Learning."* In 2024 IEEE International Conference on Big Data (Big Data) Dec 15-18, 2024.

13. Title: *"Harnessing the power of Vocal Signals in COVID-19 Detection utilizing Machine Learning*". In 2024 IEEE International Conference on Big Data (Big Data) Dec 15-18, 2024.

14. Title: "*Preparing Wearable Data for AI-powered mood and compliance prediction in HCT patients and caregivers*". In 2024 IEEE International Conference on Big Data (Big Data) Dec 15-18, 2024.

15. Title: "*Generative AI for efficiency in Higher Education and Institution Research*", Workshop at SACSCOC Annual Meeting, Dec 07-10, 2024.

16. NSF CREST/HBCU-RISE/PRP Conference, University of Texas Rio Grande Valley, Edinburgh, TX, Nov 19-20, 2024.

17. MS-CC Workshop: Campus Technology, Cybersecurity, & Research Computing Support, Alabama A&M, Oct 29-30, 2024.

18. Panelist and organizer for the NSF ExpandAI Leadership Workshop (EXAIL), SAIL Conference, Oct 7-10, 2024**.**

19. Title: "*Histopathology Imaging Data Analytics for an Integrated Model for Prostate Cancer Risk Stratification*", AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2024), Atlanta, GA, Aug 06-07, 2024.

20. Title: *"NSF Workshop to Grow Aspiring PIs in the Computer and Information Sciences in Alaska and Neighboring Regions",* Anchorage, AK, August 05-06, 2024.

21. Title: "*Virus Integration Site Detection in Tumor Genomes Using Deep Learning*", 22nd International Conference on Artificial Intelligence in Medicine (AIME 2024), AI in Oncology workshop, July 09-12, 2024.

22. Title: "*NSF Workshop on Envisioning Open Research Resources for AI*", Alexandria, VA, May 22-23, 2024.

23. Title: "*NSF Workshop on Enhancing Research Opportunities and Access to Federal Funding for Aspiring Investigators in Computer and Information Science and Engineering (CISE): Southeast*", University of Alabama, Huntsville, AL, April 07-08, 2024.

24. Title: "*AI/ML Ready mHealth and Wearables Data for Dyadic HCT*", NIH ODSS AI Supplement Program Virtual PI meeting, March 27-28, 2024.

25. Title: *"Trustworthy AI in Healthcare to advance health equity"*, Accelerate Precision Health Grantee Meeting, Chen Zuckerberg Initiative (CZI), Jan 31-Feb 2, 2024.

26. Title: "*mHealth and Wearables Data Stream Analytics for Early Insights into life-threating Diseases*", Institute of Translational Health Sciences (ITHS) seminar, University of Washington, Apr 11, 2022

27. Title: "*Big data in mHealth and wearables*", TTU Seminar, Dept. of Electrical and computer engineering, Texas Tech University, Feb 22, 2022

28. Title: "*Adaptive Normalization in Streaming Data*". ACM Conference on Big Data Research (ICBDR). Paris, France, Nov 20-22, 2019

29. Title: "*Undertaking Literature review in R*". National Wind Institute, Texas Tech University, Lubbock, TX, USA, Mar 29, 2019

30. Title: "*Unleashing the Power of Hashtags in Tweet Analytics with Distributed Framework on Apache Storm*". IEEE International Conference on Big Data, Seattle, WA, USA, Dec 10-13, 2018

31. Title: "*Harnessing the power of hashtags in tweet analytics*". IEEE International Conference on Big Data, Boston, MA, USA, Dec11-14, 2017

32. Title: "*Hepatotropic Virus Diagnosis in Human Metabolism using self-organizing map neural networks*", National conference on Applications of Data Mining in management of

metabolic and degenerative disorders, Amrita Vishwa Vidyapeetham, Mysore, India, April 16-18, 2012

State/Regional Meetings

1. Title: *"Generalizable and Interpretable Multimodal AI Fusion Framework for Precision Medicine Using Foundation Models"* , Sealy Center for Environmental Health and Medicine (SCEHM) Annual Member's Retreat 2026
   s
2. Title: *"Generalizable and Interpretable Multimodal AI Fusion Framework for Precision Medicine Using Foundation Models"*, Center for Health and Clinical Outcomes (H-COR) Research seminar series, March 4, 2026.

3. Title: *"Meet the Experts",* UTMB Data Science Consulting and Collaboration Seminar, UTMB, February 19, 2026.

4. Title: "*Multimodal AI in healthcare*", DMAC CERCIT Seminar, UTMB, January 27, 2026.

5. Title: "*Multimodal AI in healthcare*", AI Exchange talk, UTMB, November 20, 2025.

6. Title: "*AI in Education: benefits, risks, and best practices -II"*, Blackboard training, Meharry Medical College, July 25, 2024.

7. Title: "*Legal policies, regulations, and ethics in healthcare*", Homecoming and reunion, Meharry Medical College, May 16, 2024.

8. Title: "*AI in Education: benefits, risks, and best practices -I"*, Blackboard training, Meharry Medical College, May 13, 2024.

9. Title: "*Legal policies, regulations, and ethics in healthcare*", Ethical and responsible AI in medical Systems Seminar, School of Applied Computational Sciences, Meharry Medical College, Feb 27, 2024.

10. Title: "*Exploiting vocal signals and mHealth for non-invasive disease diagnostics*", 67th Annual Meharry research week, Meharry Medical College, Sep 29, 2023

11. Title: "*Digital Health Devices Data Stream Analytics for Early Insights into life-threating Diseases*", RCMI Seminar, Meharry Medical College, Dec 16, 2021

12. Research talk at University of Memphis-Meharry Program to Enhance Collaborative and Interdisciplinary Research (PECIR), April 15, 2021

13. Title: "*Applying Computing Methods to Analyze Big Genomic, EHR, and Wearables Data*", Meharry Medical College, April 06, 2021

14. Title: *"Best practices for innovative teaching",* A. Cherrie Epps, Ph.D.8th Annual School of Medicine Education Symposium, Meharry Medical College, May 11-12, 2022.

SERVICE TO THE SCIENTIFIC OR PROFESSIONAL COMMUNITY

❑ Departmental and School level at UTMB

1. SPPH Admissions committee

❑ Departmental and School level at Other Universities (Meharry Medical College)

1. **Program Coordinator, Master's in Computer Science program, 2024:** I was appointed as the program coordinator for the newly established Master's in Computer science program at SACS as per request from our Dean. In this role, I coordinated with the fellow team members to draft the program requirements, curriculum map, program goals and student learning outcomes, and syllabus document for all the courses. I took an active role to coordinate and approve the program from school for final SACSCOC approval.

2. **Member, Lab design committee, 2024:** I actively served in our Lab design committee. The goal of this committee is to provide advice on lab equipment needs, identification, and procurement. I actively contributed to this committee to build our SACS lab infrastructure.

3. **Member, Ph.D. Qualifying exam committee, 2023- Present:** I actively served in our Ph.D. qualifying exam committee member. The goal of this committee is to conduct the Ph.D. qualifying exam, design the exam, and organize logistics required. We successfully conducted the qualifying exam for our first Ph.D. cohort in 2023.

4. **Member, SACSCOC Committee, 2023-present:** I actively served in our SACSCOC committee. The goal of this committee is to collect the required paperwork and design the document for the SACSCOC approval of our programs. I am the coordinator of Master's in data science program and actively contributed to the required paperwork and currently serving in this committee to accomplish the goal.

5. **Member, Faculty Search committee, 2021 -present:** I actively served in this committee for faculty recruitment. This committee was formed in 2021 and chaired by our Dean. The goal of this committee is to review the faculty applicant profiles, conduct the interviews, and take part in the final selection of the faculty candidates. Since the start of the school. I actively contributed in most of the faculty recruitments at SACS. We started with 5 faculty members as part of this committee and so far, we have recruited 14 faculties in both the departments (computer science and data science and biomedical data science).

6. **Member, Admissions committee, 2021-present:** I actively served in this committee for the students recruitment. The goal of this committee is to review the students' profile

applied for admission, conduct the interviews, and take part in the final selection of students. Since the start of school. I actively contributed in most of the students' recruitment at SACS. We started with 4 faculty members as part of this committee and so far, we have recruited 100 students in both the departments (computer science and data science and biomedical data science).

7. **Member, Curriculum committee, 2021-present:** I actively served in this committee for curriculum development. The goal of this committee is to review the course curriculum, design syllabi for the courses, define the course schedules, approve the curricula revisions, and develop the paperwork required for program approval of SACSCOC. I actively contributed in this committee since 2021 to design the curriculum and programs including MSDS, MSBD, PhD DS, PhD BD, MSCS, MSAI, MS Cybersecurity assurance. I actively contributed to the design of 10 courses and curriculum as part of this committee.

8. **Member, Infrastructure development committee, 2021-2022**: I actively served in this committee for the infrastructure development. The goal of this committee is to provide advice on equipment needs, identification, and procurement. I actively contributed to this committee to build infrastructure for our Enterprise data analytics (EDA) division of SACS. We now have a strong high performance computing infrastructure.

9. **Member, RCMI Data Science Task Force, 2021-2022:** I actively served in this committee for the RCMI supplement award. The goal of this committee was to design all the project activities of the funded RCMI supplement award. I actively contributed in coordinating RCMI seminar series, trainings, and workshops as part of this award.

10. **Member, Objective Assessment Task Force, 2021-2022:** I actively served in this committee for the school's course objective assessment. This committee was formed in 2021. The goal of the committee was to design the assessment metrics to enhance student learning and faculty teaching and to enhance academic programs.

- ❑ University level

1. **Parliamentarian, SACS Faculty Council, Faculty senate, 2024-present:** I actively served as a parliamentarian for our university wide Faculty senate. The goal of the committee is to actively contribute to our SACS faculty council meeting and take part in the faculty senate.

2. **Member, Faculty Development CoP committee, 2021-2022:** I actively served in this committee for the university wide Faculty development CoP committee. This committee was formed in 2021. The goal of the committee was to develop a comprehensive faculty development plan for the university. I was chosen as representative for this committee from SACS and played critical role in reviewing the faculty development course and take part in discussions.

❑ State/Regional/Local

1. Organizing committee member of NSF Expand AI leadership Workshop (Co-located with SAIL 2024), organized in October 2024 in Pittsburgh, USA.

2. Planning and organizing committee member of NIH R25 summer data science bootcamp, organized from June 2023 and June 2024.

3. Planning and organizing committee member of NASA Collaborative interactive Data Science Academy, organized from July 15-19, 2024.

4. Planning and organizing committee member of NASA Meharry Data Science Academy, organized from July 17-19, 2022.

5. Planning and organizing committee member of 69th Meharry Annual Research Day.

❑ Editorial Board Positions

2026	Early career editorial board member for Journal Bioengineering
2025- Present	Guest editor for Journal Sensors special issue
2023 - Present	Guest editor for Journal Mathematics special issue
2022 – Present	Reviewer Board member for Journal Big data and Cognitive Computing
2022 - Present	Review Editor for Journal Frontiers in Big Data: Data Mining and Management Section
2020 - 2022	Review Editor for Journal Frontiers in Psychology: The Quantitative Psychology and Measurement Section

❑ Journal Service (Reviewer)

- Reviewer, Engineering Applications of Artificial Intelligence
- Reviewer, JMIR Journal of Participatory Medicine
- Reviewer, Nature Communication Medicine
- Reviewer, Health Data Science
- Reviewer, MAKE Journal
- Reviewer, Informatics Journal
- Reviewer, Medical Decision Making
- Reviewer, Journal of Personalized Medicine
- Reviewer, Automatika
- Reviewer, Diagnostics
- Reviewer, Smart Cities
- Reviewer, Clinical and Translational Medicine

- Reviewer, ACM Transactions on Knowledge Discovery from Data
- Reviewer, British Journal of Hematology
- Reviewer, Journal of Medical Internet Research
- Reviewer, IEEE Access
- Reviewer, Nursing Open Journal
- Reviewer, Electronics
- Reviewer, Frontiers in Psychology
- Reviewer, Sustainability
- Reviewer, International Journal of Environmental Research and Public Health
- Reviewer, Journal of Imaging
- Reviewer, Applied Sciences
- Reviewer, International Journal of Geo-Information
- Reviewer, Information Processing and Management
- Reviewer, Security and Communication Networks
- Reviewer, Big Data and Cognitive Computing
- Reviewer, Healthcare
- Reviewer, JMIRx Med
- Reviewer, Nature Computational Sciences
- Reviewer, Discover AI
- Reviewer, Nature Scientific Reports

❑ Reviewer for Funding Agencies or Grant

- NSF Expand AI CAP – 2024
- NSF CISE Research Expansion D-CPS/SaTC/Others – 2024
- NSF IUSE – 2024
- NSF GRFP – 2023
- NSF PIPP – 2022

❑ Conference abstract reviewer

- RCMI Consortium National Conference, Bethesda MD - 2024
- 67th Annual Student Research Day, MMC Nashville TN - 2023

LEADERSHIP AND PROFESSIONAL DEVELOPMENT TRAINING

June 23rd, 2026	Disseminate Scholarship of Teaching: How to Publish a Teaching Tool or Curriculum.
Feb 23rd, 2026	SPPH Faculty Development
Dec 4th, 2025 - Present	SPPH Grant Incubator Program (GIP)
Nov 14th, 2025	Online Teaching Retreat
Oct 20th, 2025 - Present	SPPH Writing Accountability Groups (WAGs)
Dec 9th, 2025 – Jan 20th, 2026	Grant Writing Workshop, Office of Faculty Affairs
Dec 5th, 2025	Inaugural UTMB AI Centre retreat
Jun 7th – 9th, 2024	The Teaching Professor Conference
Jun 7th – 9th, 2023	The Teaching Professor Conference
May 12th, 2022	Cherrie Epps, Ph.D.8th Annual School of Medicine Education Symposium
Apr 21st, 2022	Blackboard Digital teaching symposium

Apr 4th, 2022	NSF CISE CAREER Workshop
Apr 4th, 2022	Medical Data science seminar, University of Washington
Mar 31st, 2022	NIH AIM-AHEAD Brown Bag webinar
Oct 18th -19th, 2021	NTC Analytics Summit
Jun 7th – Sep 30th, 2021	The Teaching Professors Virtual Conference
May 3rd – 14th, 2021	IBM Skills Academy, Train-the-Trainer Workshop (Data Science)
Mar 1st, 2021	NIH Grant writing workshop
Feb 22nd, 2021	Training on Conflict of Interest, HIPAA Awareness
Feb 16th – Feb 17th, 2021	NSF Workshop on Predictive Intelligence for Pandemic Prevention

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

2015 – Present	Institute of Electrical and Electronics Engineers (IEEE)
2015 -Present	Association for Computing Machinery (ACM)

RESEARCH ACTIVITIES

My research program focuses on developing multimodal, reliable, equitable, and transparent Artificial Intelligence (AI) methods and tools to translate healthcare data into actionable insights, improving patient outcomes across various diseases. I am driven by the transformative potential of AI in healthcare, especially its ability to enhance diagnostics, treatment, and patient outcomes, while ensuring its responsible and secure implementation. Within machine learning, I am particularly interested to apply machine learning to analyze multimodal, longitudinal data streams from digital health sources (like mHealth apps and wearables) to enable chronic Disease management, Mental Health, and Cancer Support.

My research agenda has centered around the following four broad areas:

- Personalized and secure mHealth & Wearables interventions for chronic disease management, mental health, and cancer support.
- Enhancing the quality, and fairness of healthcare data
- Multimodal AI for enhanced diagnostics
- Advancing user-centered design, tools development, and applied AI research

Personalized and secure mHealth & Wearables interventions for chronic disease management, mental health, and cancer support

My area of focus in this area is to design, develop, and evaluate innovative mobile health (mHealth) solutions coupled with wearables to address critical healthcare challenges such as chronic disease management, mental health, and cancer support. After joining SACS in Spring 2021, I built my mHealth wearables sensors lab to work in this area. However, my work was initially started in collaboration with my Michigan medicine collaborators. My work on the evaluation of digital health technology-based interventions for cancer support is done on caregivers of patients undergoing hematopoietic cell transplantation (HCT). This is done using the data from the National Caregiver Health Survey, which is provided to caregivers of HCT patients. It examines the association between caregiving characteristics (i.e., care partner–recipient relationship, age, coping style, and different types of digital health technologies) and

different types of digital health technologies (i.e., smartwatch, fitness trackers, health apps). This study was published in journal *digital health* as listed below. My contribution to this work was primarily in the statistical analysis of data collected in the survey. Building on this work, in a related study, we explored whether family caregivers of HCT patients, facing emotional and psychological strain, would use digitally guided, psychology-based (PPB) activities to improve their well-being. This study was co-authored by me and published in journal *Health Psychology open* as listed below. Based on the findings of both studies, the research team designed an outpatient version of mHealth app developed for caregivers and patients undergoing HCT and implemented in a clinical trial at University of Michigan hospital.

Furthermore, to extend my work on secure wearables interventions, I worked on the research to develop methods to enable biometric authentication using smartphone fingertip Photoplethysmography (PPG) signals. This research resulted in two journal publications and in collaboration with Texas Tech University electrical engineering department. Finally, I explored the potential of leveraging human voice for disease diagnosis. In this research, we employed machine learning methods to diagnose COVID-19 by analyzing cough and breathing sounds. For the proof-of-concept, we utilized a publicly available dataset of cough and breathing sounds from COVID-infected individuals, collected through a mobile app. We successfully accomplished this work, resulting in a publication in the IEEE Big Data Conference refereed proceedings and an abstract presented at Meharry Research Week, where it won the second-best poster award. I am currently expanding this work to include a large-scale dataset requested from the University of Oxford and exploring the potential of vocal signals to detect chronic conditions, such as diabetes or pre-diabetic states, at an early stage.

In summary, under this research area of focus, I have published ***five*** journal papers, ***one*** paper in refereed conference proceedings, and ***one*** abstract. Among these, I am the lead author on one journal paper, co-authored four journal papers, and mentored one student who published a conference paper and abstract. I have also mentored a Masters' and Ph.D. student during the course of this research.

Enhancing the quality and fairness of healthcare data

AI and machine learning (ML) have revolutionized healthcare, transforming the way medical diagnoses, treatments, and patient care are approached. Despite the growing abundance of data in the healthcare sector, the inherent messiness, incompleteness, and biases within this data significantly hinder the development and deployment of effective and reliable AI solutions. Existing clinical AI methods have algorithmic and societal biases present in the development pipeline that led to misinformation, misdiagnosis and health disparities. *To address aforementioned issues, my research in this area focuses on developing novel data cleaning and preprocessing methods, as well as identifying and mitigating biases, to improve the quality of healthcare data.*

My past research has focused on developing preprocessing techniques for big data streams that are adaptive, autonomous, and scalable. For example, I have developed novel preprocessing pipelines to enhance data quality of numerical sensor data. In particular, I designed a real-time data normalization algorithm using a fixed-size sliding window to detect data evolution and

dynamically adjust normalization parameters. This work, presented at *International conference on Big Data Research*, has demonstrated significant improvements in the accuracy and reliability of downstream AI models. Furthermore, I have developed an online preprocessing algorithm for efficient and relevant textual data retrieval from major social media platforms such as Facebook and X (formerly Twitter). These approaches has been implemented in distributed framework *Apache Storm* which processes millions of posts (formerly tweets) in real-time with minimum latency. This research, published in computing journal *Information,* has shown promising results in tweet retrieval as compared to state-of-the-art methods. Both of the methods are now being used by several research groups in the computing community.

Building on my previously developed methods for enhancing data quality, I have identified their application in healthcare data where it is a bigger challenge to apply. We first explored the area of mHealth and wearable sensors data in cancer care. The physiological data collected from wearables (i.e., smartwatch) is highly unstructured often containing missing values, outliers, redundant features, and erroneous measurements leading to false conclusions/prediction. Therefore, enhancing data quality is essential for deriving meaningful insights. My research in this area has focused on developing novel methods to enrich the quality of data, captured from smartphone sensors and wearable devices to make it ready for AI/ML applications in HCT research and share the developed methods and processed data for mHealth research community. Thus, I submitted a NIH supplement proposal based on this idea in collaboration with Michigan medicine, which got funded in 2023. In this project, I led the development of preprocessing methods to build high quality mHealth dataset. This research resulted in one systematic review paper focused on data preprocessing techniques for wearable sensor studies in cancer care for AI/ML readiness published in a high-impact journal *JMIR mHealth and uHealth.* Based upon the findings of the systematic review, we developed novel pre-processing methods to build a high quality, comprehensive, standardized, AI/ML ready, and clinically meaningful wearable dataset of HCT patients and caregivers. This study was co-authored by me and published in refereed conference proceedings of *IEEE Big Data Conference* as listed below. The work is also presented in special NIH/ODSS session in ISMB conference. I am currently expanding this work to integrate the data captured from wearables with electronic health records (EHR) for combined analysis.

Building on my work in data cleaning, I have explored the critical issue of bias and fairness in healthcare data. My research in this area has focused on identifying and mitigating biases in data and models that can arise from various sources, such as data collection practices, patient demographics, and clinical workflows. Two proposals got funded in this research area. One NSF proposal goal is to build capacity doing foundational and use-inspired research on trustworthy AI in healthcare and another NIH RCMI Supplement proposal goal to is to advance trustworthy and ethical AI research in healthcare. I led the development of novel methods to detect and mitigate biases related to race, gender, and socioeconomic status in healthcare data. This work is submitted to journal *Informatics in medicine unlocked* which is under review, has demonstrated the importance of considering fairness in the development of clinical AI systems. I also organized multiple workshops and training activities to disseminate awareness about ethical AI issues and data quality in healthcare.

In summary, under this research area of focus, I have published ***one*** journal papers, ***one*** paper in refereed conference proceedings, and ***one*** abstract. ***One*** journal article is under review and ***one*** review article is under preparation. Among these, I am the lead author on one journal paper under review, co-authored two journal papers, one conference publication, and one abstract. I have mentored two Masters' students and two postdoctoral fellows during the course of this research.

Multimodal AI for Enhanced Diagnostics

Multimodal AI is revolutionizing healthcare by integrating and analyzing information from multiple sources. This integrated approach allows for more accurate diagnoses, personalized treatments, and improved patient outcomes. *My work on multimodal AI for enhanced diagnostics has focused on developing methods that effectively integrate diverse data sources, such as medical images, electronic health records (EHRs), genomics and clinical data, for improved diagnosis.* In a project funded by NIH AIM-AHEAD, I am leading the development of multimodal framework to integrate histopathology images, with clinical and genomics information for prostate cancer risk prediction. We are currently working to develop a novel deep learning framework that combines the features extracted from histopathology images with the sequencing data and train a combined model for the prostate cancer risk prediction. Furthermore the performance of multimodal will be compared with the unimodal trained on individual modalities. This research has potential to improve the diagnostic accuracy of prostate cancer risk compared to single-modality approaches and will be further extended in a clinical setting by collaborating with a medical center. We have presented the preliminary findings of this work through three abstracts at national conferences as listed below. My future research in this area will focus on developing methods for synergistic integration of multiple data modalities, and exploring multimodal fairness for enhanced diagnostics of different disease conditions. I believe these methods have potential to develop truly impactful AI solutions for healthcare.

In summary, under this research area of focus, I have published ***fives*** abstracts and ***one*** Journal article. I have mentored two masters' students and one data scientist during the course of this research.

Advancing user-centered design, tools development, and applied AI research

The growing need to connect computing with other research disciplines necessitates user-centered design, tool development, and applied AI research. *My research in this area focuses on the development of innovative tools and applying AI techniques in diverse areas such as tourism, education, computational genomics, healthcare etc*. My work consistently emphasizes the importance of user-centered design principles, resulting in tools that are not only technically sound but also effective, accessible, and meet the real needs of their diverse users.

My past work includes developing tools and packages for educational psychology, incorporating new methods for handling missing data, and creating a platform for visualizing wearable data. I believe that the data visualization has power to extract the hidden insights from the data. Thus, my research program focuses on designing visualization tools for computational genomics and educational research. My research program addresses the critical gap in user-friendly tools for computational genomics. Genomics research is increasingly data-intensive, requiring sophisticated computational analysis. However, many biologists and clinicians lack the specialized bioinformatics expertise to effectively utilize these resources. My recent

Bioengineering publication systematically reviewed RNA-sequencing visualization techniques and tools, identifying gaps and informing scientists about their availability, advantages, and limitations. In another study, we analyzed k-12 state assessment data extracted from the institutional websites of the States of Texas and Louisiana to demonstrate the utility of data visualization combined with machine learning. In the past 4 years, my work also focused on developing tools to detect viral integration sites in tumor genomes . This work was funded by an *American Cancer Society* pilot grant. We have developed several deep learning approaches to detect viral integration sites in tumor genomes and compile them in a tool to be useful for basic scientist and bioinformaticians. This research resulted in two abstracts and one conference proceedings paper. My work in applied AI research in healthcare developed an approach to predict cardiac complications of myocardial infarction patients using machine learning and measles rash identification using Deep convolutional neural network (CNN). This resulted in two conference papers in conference proceedings.

In summary, under this research area of focus, I have published ***three*** journal papers, ***three*** conference papers in refereed conference proceedings, and ***four*** abstracts presented and published in national and international conferences. Among these, I am the lead author on three abstracts, and senior author for two journal papers. I have mentored seven students during the course of this research.

❑ Peer-Reviewed Grants

Current

DIVA-AI Laboratory: Transforming Precision Health through Generalizable Multimodal Models
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 0% effort
Source of Support: UTMB Rising STARs award
Total Award Amount: $300,000
Goal: The goal of this award is to build a DIVA-AI lab focused on developing generalizable multimodal AI models for precision medicine

NSF NAIRR Pilot Resources for Deep Learning and Programming Foundations for Data Science courses for Biostatistics and Data Science Graduate Programs
Project Number: NAIRR260131
Name of PD/PI: Vibhuti Gupta
Role on Project: PI
Source of Support: NSF
Project/Proposal Start and End Date: 05/01/2026 – 05/01/2027
Total Award Amount (including Indirect Costs): $0 (Computing resource grant)
Goal: The goal of this award is to provide resources to students for the Deep learning and programming foundations courses.

Pending

UTMB Alzheimer's Disease & Related Dementias Postbaccalaureate Research Education Program (UTMB ADRD PREP)

Name of PD/PI: Heidi Spratt
Role on Project: Senior Personnel, 2% effort
Source of Support: NIH/DHHS
Project/Proposal Start and End Date: 04/2027 – 03/2032
Total Award Amount (including Indirect Costs): $2,155,510
Goal: An education project teaching trainees on topics in Alzheimer's Disease and Related Dementias postbaccalaureate research.

Developing Trustworthy Council-of-LLMs Framework for Longitudinal Patient Journey Knowledge Graph Construction for peripheral arterial disease
Name of PD/PI: Andrew Gonzalez
Role on Project: Co-I, 10% effort
Source of Support: NIH/NHLBI
Project/Proposal Start and End Date: 04/01/2027 – 03/31/2029
Total Award Amount (including Indirect Costs): $275,000
Goal: The goal of this proposal is to develop the graph-based methods for longitudinal patient journey mapping in peripheral arterial disease.

Early Identification of Aggressive Prostate Cancer Using Longitudinal EHR Foundation Models on Flatiron Data
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 20% effort
Source of Support: ACS-Flatiron Real-World Data Impact Award
Project/Proposal Start and End Date: 01/2027 – 12/2027
Total Award Amount (including Indirect Costs): $100,000
Goal: The goal of this proposal is to develop and validate a temporally aware framework for early identification of aggressive prostate cancer using EHR foundation models on Flatiron data.

Generalizable and Interpretable Multimodal Foundation Model for Predicting Aggressive Prostate Cancer Risk
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 20% effort
Source of Support: American Cancer Society (ACS)
Project/Proposal Start and End Date: 04/01/2027 – 03/31/2031
Total Award Amount (including Indirect Costs): $943,501
Goal: The goal of this proposal is to develop a generalizable and interpretable multimodal foundation model for aggressive prostate cancer risk predictions.

Light based Intraoral Imaging Platform for Detection of Oral Neoplasia and Biopsy Guidance
Name of PD/PI: Garcie Vargas
Role on Project: Co-I, 12.5% effort
Source of Support: CPRIT
Project/Proposal Start and End Date: 12/2026 – 11/2029
Total Award Amount (including Indirect Costs): $900,000

Goal: The goal of this proposal is to evaluate individual and combined components of the multimodal instrument in clinical samples and define image acquisition, visualization, and analysis workflows that may be applied in the clinical environment.

Generalizable and Interpretable Multimodal Foundation Model for Predicting Lethal Prostate Cancer Risk
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 30% effort
Source of Support: Prostate Cancer Foundation (PCF)
Project/Proposal Start and End Date: 09/2026 – 08/2028
Total Award Amount (including Indirect Costs): $1000,000
Goal: The goal of this proposal is to develop a generalizable and interpretable multimodal foundation model for Prostate cancer metastasis risk predictions.

Translational AI Loops (TRAILS) Framework: From Multimodal Patient Data to Community Impact
Name of PD/PI: Suresh Bhavnani
Role on Project: Co-I, 15% effort
Source of Support: NIH/NCATS
Project/Proposal Start and End Date: 04/2027 – 04/2032
Total Award Amount (including Indirect Costs): $2,500,000
Goal: The goal of this proposal is to develop multimodal patient data for community impact.

Past

Data Management and Analysis Core for Comparative Effectiveness Research on Cancer in Texas
Name of PD/PI: Yong-Fang Kuo
Role on Project: Co-I, 20% effort
Source of Support: CPRIT
Project/Proposal Start and End Date: 09/2023 – 08/2026
Total Award Amount: $2,936,731
Goal: The goal of the Core is to serve as a hub for cancer comparative effectiveness research in Texas.

Multimodal Hybrid-AI (MoHAI): Bridging Data Complexity and Interpretability for Predicting Depression Risk in Alzheimer's Disease
Name of PD/PI: Vibhuti Gupta
Role on Project: MPI, 0% effort
Source of Support: UTMB Office of Team Science
Project/Proposal Start and End Date: 04/01/2026 – 08/14/2026
Total Award Amount: $50,000
Goal: The goal of this award is to develop an interpretable multimodal framework using foundation models to predict depression risk in Alzheimer's disease.

NSF Conference: NAIRR Pilot Expansion: FA1: Mid-South Conferences on Cyberinfrastructure Advances to Enable Interdisciplinary AI Research in Health

Project Number: 2526364
Name of PD/PI: Li Qian
Role on Project: Co-PI, 3% effort
Source of Support: NSF
Project/Proposal Start and End Date: 07/01/2025 – 12/31/2026
Total Award Amount (including Indirect Costs): $125,133
Subaward Amount (including Indirect Costs): $28,082
Goal: The goal of this proposal is to provide training, guidance, and support to the formation and furtherance of interdisciplinary research at the nexus of Advanced cyberinfrastructure, health disparities, and domain sciences.

NSF NAIRR Pilot Resources for Machine Learning, Programming, and Big data courses for Data Science and Biomedical Data Science Graduate Programs
Project Number: NAIRR240340
Name of PD/PI: Vibhuti Gupta
Role on Project: PI
Source of Support: NSF
Project/Proposal Start and End Date: 03/12/2025 – 03/11/2026
Total Award Amount (including Indirect Costs): $0 (Computing resource grant)
Goal: The goal of this award is to provide resources to students for the Machine learning, programming, and Big data courses.

Analyzing large scale histopathology images from prostate cancer for multimodal data integration
Project Number: **CNMS2025-R-03166**
Name of PD/PI: Vibhuti Gupta
Role on Project: PI
Source of Support: Oak Ridge
Project/Proposal Start and End Date: 04/14/2025 – 07/31/2025
Total Award Amount (including Indirect Costs): $0 (Computing resource grant)
Goal: The goal of this award is to provide oak ridge computing resources to analyze large scale histopathology images using advanced deep learning techniques.

NSF- Excellence in Research: A Trustworthy and Privacy-preserving Framework for Latency-sensitive Video-enabled Health Incident-Response Applications
Project Number: 2401928
Name of PD/PI: Uttam Ghosh
Role on Project: Co-PI, 4% effort
Source of Support: NSF
Project/Proposal Start and End Date: 09/01/24 – 08/31/27
Total Award Amount (including Indirect Costs): $611,959
Goal: The goal of this proposal is to develop trustworthy and privacy-preserving framework for video enabled health incident response applications.

NSF-Conference: EXAIL: NSF Expand AI Leadership Workshop
Project Number: 2437262

Name of PD/PI: Dhireesha Kudithipudi
Role on Project: Co-PI, 1.5% effort
Source of Support: NSF
Project/Proposal Start and End Date: 09/1/24 – 08/31/25
Total Award Amount (including Indirect Costs): $137,071
Goal: The goal of this proposal is to conduct a conference workshop for all the Expand AI awardees in SAIL 2024 Conference.
Outcome: Organized the NSF Expand AI leadership workshop and participated in a panel discussion on Insights from CAP Awardees: Lessons Learned and Best Practices

NSF CAP: Capacity Buildings for Trustworthy AI in Medical Systems (TAIMS)
Project Number: 2334391
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 13% effort
Source of Support: NSF
Project/Proposal Start and End Date: 10/01/23 – 09/30/25
Total Award Amount (including Indirect Costs): $395,703
Goal: The goal of this planning award is to build capacity for a potential AI institute at Meharry for doing foundational and use-inspired research on trustworthy AI in medical systems.
Outcome: Organized multiple training and webinar series for ethical AI training and published 3 conference papers and 2 journal articles with one under review. Mentored 6 students.

Unified and Modular Framework for Multimodal Data Integration with AI/ML Models to Address Prostate Cancer Disparities
Project Number: 3OT2OD032581-01S1
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 30% effort
Source of Support: NIH
Project/Proposal Start and End Date: 09/17/23 – 09/16/25
Total Award Amount (including Indirect Costs): $1,267,364
Goal: The objective of this consortium development project is to develop a unified and modular framework to systematically integrate clinical, genomics, and imaging data, leveraging AI/ML techniques to computationally predict various PCa risk groups.
Outcome: Published 2 abstracts and presented poster. 2 Journal articles are in preparation. Mentored 3 students.

The RCMI Program in Health Disparities Research at Meharry Medical College – Supplement- Ethical Considerations in AI/ML
Project Number: 5U54MD007586-37S3
Contact PI: Samuel E. Adunyah
Supplemental PI: Vibhuti Gupta
Role on Project: MPI, 8% effort
Source of Support: NIMHD
Project/Proposal Start and End Date: 08/1/23 – 05/31/24
Total Award Amount (including Indirect Costs): $218,250
Goal: The goal of this award is to advance trustworthy and ethical AI in healthcare.

Outcome: Organized multiple training and webinar series for ethical AI training within the RCMI community and 2 journal articles are under review. Mentored 2 students.

The Collaborative interactive Data Science Summer Academy
Project Number: 23-MUREP PSI-0005
Name of PD/PI: Eugene Levin
Role on Project: Co-I, 2.5% effort
Source of Support: NASA
Project/Proposal Start and End Date: 06/01/23–05/31/28
Total Award Amount (including Indirect Costs): $418,448
Goal: The key objective of the program is to stimulate curiosity in the cross-cutting field of data science through data science-driven, hands-on, STEM activities deploying real-life application scenarios and industrial grade robotic systems.
Outcome: Organized sessions for k-12 students to understand data science and programming in summer.

Patient Oriented Research and Mentoring in Hematopoietic Cell Transplantation- NHLBI Supplement
Project Number: 3K24HL156896-03S1/SUBK0019815
Name of PD/PI: Sung Won Choi
Role on Project: MPI, 17% effort
Source of Support: NIH
Project/Proposal Start and End Date: 08/1/23 – 07/31/24
Total Award Amount (including Indirect Costs): $114,123
Subaward Amount (including Indirect Costs): $54,543
Goal: The goal of this supplement award is to develop high quality mHealth dataset generated from smartphones and wearable sensors.
Outcome: Published 1 abstract, 1 conference article and 1 journal article. Mentored 2 postdoctoral fellows.

The RCMI Program in Health Disparities Research at Meharry Medical College – Supplement- Data Science Subaward
Project Number: 3U54MD007586-35S5
Name of PD/PI: Samuel E. Adunyah
Subaward PI: Zhenbang Chen
Role on Project: Co-PI
Source of Support: NIMHD
Project/Proposal Start and End Date: 01/15/23 - 06/30/23
Total Award Amount (including Indirect Costs): $15,000
Goal: This subaward is proposed to identify the role of circRNAs in prostate cancer malignancy and disparities using Big data analytics techniques.
Outcome: Published 1 abstract

ACS DICRIDG Pilot Award
Project Number: DICRIDG-21-071-01-DICRIDG
Contact PI: Samuel E. Adunyah
Sub Project PI: Vibhuti Gupta

Role on Project: MPI
Source of Support: ACS
Project/Proposal Start and End Date: 07/01/2021-06/30/2024
Total Award Amount: $45,000
Goal: The goal of this project is to detect virus integration sites in tumor genomes for the viruses responsible for cancer using deep learning techniques.
Outcome: Published 2 abstracts and 1 conference paper. Mentored 2 students

The RCMI Program in Health Disparities Research at Meharry Medical College – Supplement
Project Number: 3U54MD007586-35S5
Name of PD/PI: Samuel E. Adunyah
Supplemental PI: Qingguo Wang
Role on Project: Co-I
Source of Support: NIMHD
Project/Proposal Start and End Date: 09/01/21 - 06/30/23
Total Award Amount (including Indirect Costs): $363,750
Goal: This administrative supplement is purposed to help the data scientists at Meharry Medical College understand and address the computing needs of the RCMI (The Research Centers in Minority Institutions) researchers at Meharry, substantially enhance their data analytics skills, and foster collaborations with them.
Outcome: Participated and organized multiple training sessions. Co-authored 2 journals articles.

MUREP Precollege Summer Institute
Project Number: NNH23ZHA001N-MUREPPSI
Name of PD/PI: Eugene Levin
Role on Project: Co-I, 2.5% effort
Source of Support: NASA
Project/Proposal Start and End Date: 04/29/22 - 08/30/23
Total Award Amount (including Indirect Costs): $65,000
Goal: The key objective of the program is to stimulate curiosity in the cross-cutting field of data science and emerging technologies through a discovery-based summer immersion program. Program activities will build statistical and critical thinking skills while inspiring and diversifying the next generation of explorers, researchers, and data scientists.
Outcome: Organized sessions for k-12 students to understand data science and programming in summer.

MRI: Acquisition of a High-Performance Computer System to Support Research and Training in Computational Biology and Data Science at Meharry Medical College
Project Number: 2117282
Name of PD/PI: Aize Cao
Role on Project: Senior Key Personnel
Source of Support: NSF
Project/Proposal Start and End Date: 10/2021-09/2024
Total Award Amount (including Indirect Costs): $671,411

Goal: The goal of this project is to acquire a high-performance computing (HPC) system for the School of Applied Computational Sciences at Meharry Medical College.
Outcome: Multiple articles are part of this project in using HPC facilities.

Declined

Causal, Logic-grounded AI for Interpretable Clinical Reasoning (CLAIRE)
Name of PD/PI: Subash Neupane
Role on Project: Co-PI, 5% effort
Source of Support: NSF
Project/Proposal Start and End Date: 08/2026 – 07/2029
Total Award Amount (including Indirect Costs): $530,762
Subaward Amount (including Indirect Costs): $174,982
Goal: The goal of this proposal is to develop novel Neuro-Symbolic AI methods for clinical decision support.

Agentic AI for Trustworthy Clinical Reasoning and Adaptive Care Coordination
Name of PD/PI: Subash Neupane
Role on Project: Co-PI, 10% effort
Source of Support: Amazon Research Award
Project/Proposal Start and End Date: 02/2026 – 02/2027
Total Award Amount (including Indirect Costs): $70,000
Subaward Amount (including Indirect Costs): $17,500
Goal: This project advances Agentic AI in Healthcare through the development of CLAIRE-Health, an open source AWS native framework for trustworthy multi-agent reasoning and adaptive care coordination.

Generalizable and Interpretable Multimodal AI Fusion Framework for Precision Oncology
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 0% effort
Source of Support: Blavatnik Foundation award for young scientists
Total Award Amount (including Indirect Costs):
Goal: The goal of this award is to provide support for young investigators for expanding the innovative research

AI-Driven Multimodal Modeling to Optimize Targeted Interventions for Fatigue & Altered Cognition
Name of PD/PI: Cynthia Li
Role on Project: Co-I, 0% effort
Source of Support: UTMB AXIS Pilot
Total Award Amount: $50,000
Goal: The goal of this pilot project is to develop and apply an artificial intelligence (AI) framework/tool that integrates multimodal sources of data from clinical laboratory, patient-reported survey data, and biospecimens to identify patterns associated with Brain Injury Associated Fatigue & Altered Cognition and generate preliminary evidence for mechanistic pathways and risk stratification to inform targeted treatments.

Improving Quality of Care for Older Adults Living with Opioid Use Disorder
Name of PD/PI: Yong Fang Kuo
Role on Project: Co-I, 15% effort
Source of Support: NIH NIAD
Project/Proposal Start and End Date:
Total Award Amount (including Indirect Costs):
Goal: The goal of this proposal is to understand the effectiveness and toxicity of MOUD in older adults with opioid use disorder.

RAPP: From Human-Centered AI to the Targeted Use of Community and Government Safety Net Resources
Name of PD/PI: Suresh Bhavnani
Role on Project: Co-I, 15% effort
Source of Support: PBIF
Project/Proposal Start and End Date: 11/2025 – 11/2027
Total Award Amount (including Indirect Costs): $1,190,732
Goal: The goal of this proposal is to develop Human centered AI for community

Re-imagining the Translation of Scientific Evidence into Health Policy: The Collaborative AI for Policy- Design Framework
Name of PD/PI: Suresh Bhavnani
Role on Project: Co-I, 10% effort
Source of Support: Texas Academy of Medicine, Engineering, Science
Project/Proposal Start and End Date: 03/2026 – 02/2028
Total Award Amount (including Indirect Costs): $500,000
Goal: The goal of this proposal is to develop a framework to translate scientific evidence into public health policies

Development and Evaluation of Multimodal Framework for Longitudinal Prediction of Alzheimer's Disease Progression
Name of PD/PI: Vibhuti Gupta
Role on Project: PI, 0% effort
Source of Support: Claude D. Pepper OAIC Pilot Proposal
Project/Proposal Start and End Date: 09/01/2026 – 08/31/2027
Total Award Amount (including Indirect Costs): $50,000
Goal: The objective of this pilot proposal is to develop a multimodal framework that jointly learns from multimodal data to predict longitudinal Alzheimer's Disease (AD) progression (i.e., transition along AD continuum stages, time to AD Dementia) and to evaluate for biological validity, generalizability, and interpretability.

BIBLIOGRAPHY:
63 peer-reviewed journal articles, conference papers, and abstracts.
Google Scholar Citations
Citations: 884; h-index: 15; i10-index: 19 (10/1/2026).

Published

❑ Articles in Peer-reviewed Journals
*** (equal/first/corresponding author)**

1. Taku, E. M., **Gupta, V.***, & Singhal, A. (2026). Personalized Type 1 Diabetes Management: Reinforcement Learning–Based Insulin Dosing and Glucose Forecasting. *JMIR diabetes*, *11*, e79195. DOI: https://doi.org/10.2196/79195

2. Tiwari A. & **Gupta, V.*** (2026). A memory fabric for conversational AI agents enabling shared and persistent multiuser memory. *Discover Artificial Intelligence*, *6, 239*. DOI: https://doi.org/10.1007/s44163-026-00992-z

3. Salahuddin, M. F., Walker, J., Zambrana, E. H., **Gupta, V.**, Jung, K., Pandi-Perumal, S. R., & Manzar, M. D. (2025). Self-Regulation Mediates the Relationship Between Stress and Quality of Life in Shift-Working Healthcare Professionals: Behavioral Clustering Insights. *European Journal of Investigation in Health, Psychology and Education*, *15*(9), 180. DOI: https://doi.org/10.3390/ejihpe15090180

4. De Matas, J., Wang, J., & **Gupta, V.** (2025). Narrative Transparency in AI-Driven Consent. *The American Journal of Bioethics*, *25*(4), 136-138, PMID: 40192675 DOI: https://doi.org/10.1080/15265161.2025.2470675

5. **Gupta, V**.*, Broughton, J., Rukundo, A., and Pinky, L. (2025), Learning Unbiased Risk Prediction Based Algorithms in Healthcare: A case study with primary care patients. In *Informatics in Medicine Unlocked Journal, 54, 101627.* DOI: https://doi.org/10.1016/j.imu.2025.101627

6. Manzoor, F., Tsurgeon, C. A., & **Gupta, V.*** (2025). Exploring RNA-Seq Data Analysis Through Visualization Techniques and Tools: A Systematic Review of Opportunities and Limitations for Clinical Applications. *Bioengineering*, *12*(1), 56, PMID: 39851330. DOI: https://doi.org/10.3390/bioengineering12010056.

7. Ortiz, B. L., **Gupta, V.***, Kumar, R., Jalin, A., Cao, X., Ziegenbein, C., Singhal, A., and Sung, W.C. (2024), Data Preprocessing Techniques for Artificial Intelligence (AI)/Machine Learning (ML)-Readiness: Systematic Review of Wearable Sensor Data in Cancer Care. *JMIR mHealth and uHealth*, 12(1), PMID: 38626290 . DOI: 10.2196/59587

8. Taylor, L., **Gupta, V.***, & Jung, K. (2024). Leveraging Visualization and Machine Learning Techniques in Education: A Case Study of K-12 State Assessment Data. *Multimodal Technologies and Interaction*, 8(4), 28. DOI: https://doi.org/10.3390/mti8040028

9. Wang, Q., **Gupta, V.**, Cao, A., Gangula, P., Xie, H., Singh, R., Gary, T., Adunyah, S., Aramandla,R., Shanker, A. (2024), Enhancing Data Science and Genomics Capacity of a Historically Black Medical College Through Interdisciplinary Training and Research Collaborations. In *Journal of Biotechnology and Biomedicine*, 7, 425-432, PMID: 39735691. DOI: 10.26502/jbb.2642-91280166.

10. Wang, Q., **Gupta, V.**, Cao, A., Singhal, A., Gary, T., & Adunyah, S. E. (2023). A Case Study of Enhancing the Data Science Capacity of an RCMI Program at a Historically Black Medical College. *International Journal of Environmental Research and Public Health*, 20(6), 4775, PMID: 36981686. DOI: 10.3390/ijerph20064775

11. Hoodin, F., **Gupta, V.**, Mazzoli, A., Braun, T., & Choi, S. W. (2023). Positive psychology interventions for family caregivers coping with cancer: Who will use them? *Health Psychology Open*, 10(2), PMID: 38152307. DOI: https://doi.org/10.1177/20551029231224358

12. **Gupta, V.** Post-transplant dynamic risk prediction (2022). *Nature Computational Sciences*, 2, 144–145, PMID: 38177450. DOI:10.1038/s43588-022-00220-5

13. Raj M, **Gupta, V.***, Hoodin F, Clingan C, Roslin C, Yahng L, Braun T, Choi SW. (2022), Evaluating mobile Health technology use among cancer caregivers in the digital era. *Digital Health*. Jun; 8:20552076221109071, PMID: 35769358. DOI: https://doi.org/10.1177/20552076221109071

14. Ortiz, B. L., **Gupta, V.**, Chong, J. W., Jung, K., & Dallas, T. (2022) User Authentication Recognition Process Using Long Short-Term Memory Model. *Multimodal Technologies and Interaction*, 6(12), 107. DOI: https://doi.org/10.3390/mti6120107

15. Ortiz, B. L., Chong, J. W., **Gupta, V.,** Shoushan, M., Jung, K., & Dallas, T. (2022) A Biometric Authentication Technique Using Smartphone Fingertip Photoplethysmography Signals. *IEEE Sensors Journal*, 22(14), 14237-14249. DOI: http://dx.doi.org/10.1109/JSEN.2022.3176248

16. Nguyen, V. T., Jung, K., **Gupta, V.** Examining data visualization pitfalls in scientific publications. (2021) *Visual Computing for Industry, Biomedicine, and Art Journal*, *Springer Nature* 4(1), 1-15, PMID: 34714412. DOI: https://doi.org/10.1186/s42492-021-00092-y

17. Weyand, A. C., Fitzgerald, K. D., McGrath, M., **Gupta, V.**, Braun, T., Quint, E.H., & Choi, S. W. (2021) Depression in adolescent females with heavy menstrual bleeding, *Journal of Pediatrics*, 240, 171-176, PMID: 34517012. DOI: https://doi.org/10.1016/j.jpeds.2021.09.007

18. Flora, C., Tyler, J., Mayer, C., Warner, D., Khan, S., **Gupta, V.**, Lindstrom, R., Mazzoli, A., Rozwadowski, M., Braun, M., Ghosh, M., Forger, D., Choi, S. W., Tewari, M. (2021) High-frequency temperature monitoring and computational analysis enables early detection of febrile adverse events in hematologic cancer patients, *Cancer Cell,* 39(9), 1167-1168, PMID: 34388378. DOI: https://doi.org/10.1016/j.ccell.2021.07.019

19. **Gupta, V.***, Raj, M., Hoodin, F., Lilian, Y., Braun, T. & Choi, S. W. (2021) Electronic Health Record Portal Use by Family Caregivers of patients undergoing Hematopoietic

Cell Transplantation: A National Survey Study, *JMIR Cancer, 7(1),* PMID: 33687332. DOI: https://doi.org/10.2196/26509

20. **Gupta, V.***, Braun, T. M., Chowdhury, M., Tewari, M., & Choi, S. W. (2020) A Systematic Review of Machine Learning Techniques in Hematopoietic Stem Cell Transplantation (HSCT). *Sensors*, 20(21), 6100, PMID: 33120974. DOI: https://doi.org/10.3390/s20216100

21. **Gupta, V.***, Jung, K., Yoo, S. (2020) Exploring the Power of Multimodal Features for Predicting the Popularity of Social Media Image in Destination Marketing, *Multimodal Technologies & Interaction*, 4(3), 64. DOI: https://doi.org/10.3390/mti4030064

22. **Gupta, V.***, & R. Hewett. (2020) Real-time Tweet Analytics using Hybrid Hashtags on Twitter Big Data, *Information*, 11(7), 341. DOI: https://doi.org/10.3390/info11070341

23. Jung, K., Lee, J., **Gupta, V.** & Cho, G. (2019) Comparison of Bootstrap Confidence Interval Methods for GSCA Using Monte Carlo Simulations. *Frontiers in psychology*, 10, 2215, PMID: 31681066. DOI: https://doi.org/10.3389/fpsyg.2019.02215

The following are peer-reviewed publications in the proceedings of IEEE Big Data, ACM CHASE and ACM CAIS Conference. With an acceptance rate of 20% and a Google Scholar h5-Index=54, this conference is a leading archival source for publications in the field of big data and machine learning, a subfield of computer science, and is therefore included in this section. The papers range from 6-8 pages.

24. Moore, C., **Gupta, V. ***, Neupane, S., & Tripathi, H. (2026). An Efficient and Interpretable Learning Approach for Large-Scale Histopathology Data. In *2026 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE),* Aug 03-06 (pp. 541-546). IEEE. DOI: 10.1109/CHASE69719.2026.00092

25. Pounds, D., **Gupta, V. ***, Tripathi, H., & Neupane, S. (2026). Forecasting Minute-by-Minute Stress, Anxiety, and Affective States Using Time-Series Analysis of Wearable Sensor Data. In *2026 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE),* Aug 03-06 (pp. 517-522). IEEE. DOI: 10.1109/CHASE69719.2026.00088

26. Tripathi, H., Neupane, S., Mittal. S., Rahimi, S., **Gupta, V.** (2026). Towards a HIPAA Compliant Agentic AI System in Healthcare. In 2026 *1st ACM Conference on AI and Agentic Systems (ACM CAIS 2026)*, May 26-29 (pp. 738-754). DOI: https://doi.org/10.1145/3786335.3813175

27. Baidya, S. & **Gupta, V.**, Predicting Cardiac Complications of Myocardial Infarction Patients Using Machine Learning. The REU Symposium, In 2024 *IEEE International Conference on Big Data (Big Data)* 2024 Dec 15-18 (pp. 7454-7462). DOI: https://doi.org/10.1109/BigData62323.2024.10826091

28. Williams, L. & **Gupta, V.**, A Novel Pipeline for Virus Integration Sites Detection in Tumor Genomes Using Deep Learning. In 2024 *IEEE International Conference on Big Data (Big Data)* 2024 Dec 15-18 (pp. 4946-4951). DOI: https://doi.org/10.1109/BigData62323.2024.10825037

29. Mann, A., Ajinkya, P. J., Matovu, R., **Gupta, V**., Harnessing the power of Vocal Signals in COVID-19 Detection utilizing Machine Learning. In 2024 *IEEE International Conference on Big Data (Big Data)* 2024 Dec 15-18 (pp. 4564-4570) DOI: https://doi.org/10.1109/BigData62323.2024.10825050

30. Ziegenbein, C. B., Ortiz, B.L., **Gupta, V.**, Choi S.W. Preparing Wearable Data for AI-powered mood and compliance prediction in HCT patients and caregivers. In 2024 *IEEE International Conference on Big Data (Big Data)* 2024 Dec 15-18 (pp. 4996-5005) DOI: https://doi.org/10.1109/BigData62323.2024.10825132

31. Glock K, Napier C, Gary T, **Gupta V**, Gigante J, Schaffner W, Wang Q. Measles rash identification using transfer learning and deep convolutional neural networks. In 2021 *IEEE International Conference on Big Data (Big Data)* 2021 Dec 15 (pp. 3905-3910). IEEE. DOI: https://doi.org/10.1109/BigData52589.2021.9671333

32. **Gupta, V.***, & Hewett, R. Adaptive Normalization in Streaming Data. *In Proceedings of 3rd International Conference on Big Data Research (ICBDR)*, Paris, France, Nov 2019. DOI: https://doi.org/10.1145/3372454.3372466

33. **Gupta, V.***, & Hewett, R. Unleashing the Power of Hashtags in Tweet Analytics with Distributed Framework on Apache Storm. *In Proceedings of IEEE International Conference on Big Data*, Seattle, WA, USA, Dec 2018. DOI: https://doi.org/10.1109/BigData.2018.8622302

34. **Gupta, V.***, & Hewett, R. Harnessing the power of hashtags in tweet analytics. *In Proceedings of IEEE International Conference on Big Data*, Boston, MA, USA, Dec 2017. DOI: https://doi.org/10.1109/BigData.2017.8258194

Under-review

❑ Articles in Peer-reviewed Journals

*** (equal/first/corresponding author)**

35. Wang Z., Gupta, V., Pinky, L., Mohd-Ibrahim, I., Higa, L., Feng, Z., Yang, H., Fu, Y., Chen, Z., WM Loo, L., Yu, H., Deng, Y. Multi-Fusion Artificial Intelligence Models Integrating Multi-Omics Data for Enhanced Prostate Cancer Risk Stratification Associated with Gleason Score. *Genes & Diseases*.

36. Manzoor, F., **Gupta, V.**, Pinky, L., Wang, Z., Chen, Z., Deng, Y., & Neupane, S. (2026). A Systematic Review of Multimodal Deep Learning and Machine Learning Fusion Techniques for Prostate Cancer Classification. *MDPI Cancer*.

- Peer-Reviewed Abstracts

1. Anette, G., **Gupta, V.,** & Yong-Fang Kuo. Multimodal Survival Models in Prostate Cancer Using Foundation Models, CPRIT Innovations VII: The Cancer Prevention and Research Conference, Galveston, TX, Oct 05-07, 2026.

2. Anette, G. and **Gupta, V.** Multimodal Survival Models in Cancer Using Deep Learning, UTMB Research Summit, Galveston, TX, May 07, 2026.

3. Anette, G. and **Gupta, V.** A Comparative Analysis of Survival Models on clinical datasets with high censoring and low events, Public Health Symposium, National Public Health Week, UTMB, Galveston, TX, Apr 08, 2026.

4. Moore, C., Neupane, S., **Gupta, V.**, Bhavnani, S., Wang, Z., Chen, Z., Deng, Y., Pinky, L. A Novel Efficient and Scalable Approach to Process Large-Scale Histopathology Data for Prostate Cancer, AIM-AHEAD Southeast Hub Research Symposium, Nashville, TN, Mar 26-27, 2026.

5. Moore, C., Neupane, S., **Gupta, V.**, Bhavnani, S., Wang, Z., Chen, Z., Deng, Y., Pinky, L. A Novel Generalizable Approach to Integrate Multimodal Data for Prostate Cancer, AIM-AHEAD Southeast Hub Research Symposium, Nashville, TN, Mar 26-27, 2026.

6. Bhavnani, S., Schmidt, S., Zhang, W., Pappadis, M., **Gupta, V.**, Reistetter, T., Kulesza, C., Kummerfeld, E. Causal Discovery Analysis Using Composite AI Visualizations: from NMDoH Pathways to Precision Policies, Amplify Informatics Conference, American Medical Informatics Association Annual Symposium (AMIA 2026), Denver, CO, May 18-21, 2026.

7. Bhavnani, S., Zhang, W., Pappadis, M., Schmidt, S., Wang, Z., **Gupta, V.**, Kulesza, C., Kummerfeld, E, Wang, J. The Role of Bipartite Network Models in Addressing Fairness, Performance, and Interpretability of NMDoH Predictive Models, Amplify Informatics Conference, American Medical Informatics Association Annual Symposium (AMIA 2026), Denver, CO, May 18-21, 2026.

8. Pounds, D. and **Gupta, V.** Predicting Self-Reported Stress and Affective States Using Wearable Sensor Data, American Medical Informatics Association Annual Symposium (AMIA 2025), Atlanta, GA, Nov 15-19, 2025.

9. Bengie L. Ortiz, **Gupta, V.**, and Sung, W.C. Quantifying Physiological Parameters of HCT Patients and Their Caregivers from Wearables for Post-Transplant Effects Monitoring, IEEE-EMBS International Conference on Body Sensor Networks (IEEE-EMBS BSN 2025), Los Angeles, CA, Nov 3-5, 2025.

10. Moore, C., Shaikh, F., **Gupta, V.,** Pinky, L., Wang, Z., Chen, Z., Deng, Y., and Bhavnani, S. A Novel Pipeline using UNI and CLAM for prostate cancer risk assessment for Histopathology, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, Jul 08-09, 2025.

11. Shaikh, F., Moore, C., **Gupta, V.,** Pinky, L., Wang, Z., Chen, Z., Deng, Y., and Bhavnani, S. Automated histopathological pipeline for prediction of prostate cancer risk using nuclei-based WSI tiling, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, Jul 08-09, 2025.

12. Wang, Z., **Gupta, V.,** Pinky, L., Moore, C., Fu, Y., Chen, Z., Deng, Y. Integrative Machine Learning Approaches for Predicting Prostate Cancer Risk Using Multi-Omics Data, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, Jul 08-09, 2025.

13. Bhavnani, S., Zhang, W., Golovko, G., **Gupta, V.,** Danielska, L., Wolf, S. and Finnerty, C. Modeling Burn Subtypes Using Human Centered AI, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, Jul 08-09, 2025.

14. Bhavnani, S., Zhang, W., Hunter, R., and **Gupta, V.** Predictive Modeling of Non-Medical Determinants of Health in Cancer, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2025), Dallas, TX, Jul 08-09, 2025.

15. Mann, A., **Gupta, V.,** Pinky, L., Wang, Z., Chen, Z., and Deng, Y., Histopathology Imaging Data Analytics for an Integrated Model for Prostate Cancer Risk Stratification, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2024), Atlanta, GA, Aug 06-07, 2024.

16. Wang, Z., Mann, A., Theodore, J., Fu, Y., **Gupta, V.,** Pinky, L., Chen, Z., and Deng, Y. An Integrative Multi-Omics Analysis on Prostate Cancer Risk, AI for Health Equity Symposium AIM-AHEAD Annual Meeting (AIHES 2024), Atlanta, GA, Aug 06-07, 2024.

17. Bengie L.Ortiz., Kumar, R., **Gupta, V.,** Ziegenbein, C., Aditya, J., and Sung, W.C., AI/ML Ready mHealth and Wearables Data for Dyadic HCT, NIH/ODSS special track at 32nd International Conference on Intelligent Systems for Molecular Biology (ISMB 2024), Montreal, Canada, July 12-16, 2024.

18. Williams, L. and **Gupta, V.**, Virus Integration Site Detection in Tumor Genomes Using Deep Learning, 22nd International Conference on Artificial Intelligence in Medicine (AIME 2024), AI in Oncology workshop, July 09-12, 2024.

19. **Gupta, V.**, Impact of Flipped learning and Group problem solving for improving Python programming skills in an Online Environment, Teaching Professor's conference, New Orleans, LA, USA, June 07-09, 2024.

20. Mann, A., Wang, Z., Theodore, J., Xiao, X., **Gupta, V.**, Pinky, L., Chen, Z., and Deng, Y., A Multimodal Dataset for Evaluating an Integrated Model for Prostate Cancer risk stratification, 19th Annual Texas Conference on Health Disparities, Fort Worth, TX, USA, June 06-07, 2024.

21. Mann, A. and **Gupta, V**., Harnessing the power of Vocal Biomarkers in COVID-19 Detection Utilizing Machine Learning, Meharry research week, Nashville, TN, USA, Oct 2023.

22. **Gupta, V.,** Sanders, E. and Wang, Q., Detection of Virus Integration Sites in Tumor Genomes Using Deep Convolutional Neural Networks, ACS Conference, Atlanta, GA USA, Sep 2023.

23. **Gupta, V.** and Chen, Z., Analysis of Circular RNA levels in Prostate Cancer using Databases, RCMI Conference, Bethesda, MD, USA, Apr 2023.

24. Jung, K., Lee, J., **Gupta, V.**, Kim, S., & Hwang, H. Simulation Based Investigation of Optimal Modeling Approaches for SEMs with Ordinal Variables, *The NCME Annual Meeting*, Toronto, Canada, Apr 2019.

25. Jung, K., Lee, J., **Gupta, V.**, Kim, S., & Hwang, H. Application of Generalized Structured Component Analysis for Brain Connectivity Research, *KSEA West Gulf Coast Regional Conference*, Houston, TX, USA, Oct 2018.

26. Pyakurel, A., **Gupta, V.**, Li, Y., Oyedeji, B., Gittner, K., Pitalua-Rodriguez, M., Guthrie, H., Shetty, R., Mengel, S., Khan, H. MR., Vadapalli, R., & Gittner, L. A Multidisciplinary Preliminary Approach to Systems Thinking for CVD Risk Prevention, *Academy Health National Health Policy Conference*, New Orleans, LA, USA, Jun 2017.

27. Li, Y., Mengel, S., Gittner, L., Pitalua-Rodriguez, M., **Gupta, V.**, Khan, H. MR., Gittner, K., & Pyakurel, A. Relational Analysis through Graph Data Reduction Technique, *American Public Health Association Annual Meeting & Expo (APHA)*, Atlanta, GA, USA, Nov 2017.

28. Pitalua-Rodriguez, M., Mengel, S., Gittner, L., Gittner, K., **Gupta, V.**, Pyakurel, A., Li, Y., Vadapalli, R., & Khan, H. MR. Visualizing Disease Patterns along I-20 from El Paso to Shreveport. *American Public Health Association Annual Meeting & Expo (APHA)*, Atlanta, GA, USA, Nov 2017.

29. Gittner, K., Gittner, L., Khan, H. MR, Mario Pitalua, Pitalua-Rodriguez, M., Pyakurel, A., **Gupta, V.**, & Li, Y., Contrasting Models of Cardiovascular Disease Prediction:

Reevaluating Age Adjusted Variables. *American Public Health Association Annual Meeting & Expo (APHA)*, Atlanta, GA, USA, Nov 2017.

- ❑ Other (Non-Peer Review):
  - Thesis/Dissertation

    PhD. Thesis Title: Big Data stream analytics with AI techniques, (2019), *Texas Tech University, Texas.*
  - Webinars
    - Ethical and responsible AI webinar series, 2024, Links:
      - Recordings and material: https://sacsmeharry.org/taims/#tab-seminars
        - GitHub repository link: https://github.com/vibgupta123/Ethical-and-Responsible-AI-seminar-series
    - Trustworthy AI in medical systems planning workshop, 2024, Links:
      - Recordings and material: https://sacsmeharry.org/taims/#tab-workshops

    - Public Education materials
      - Trustworthy AI teaching modules for K-12 educators' course (9 modules course), 2024, Links:
        - Webpage link: https://sacsmeharry.org/taims/#tab-education
        - GitHub repository link: https://github.com/vibgupta123/Trustworthy-AI-teaching-modules-for-High-school-educators
      - NIH R25 Summer Data Science Bootcamp, 2023, Link:
        - Recordings and material: https://sites.google.com/view/nih-r25-talk/home
        - GitHub repository link: https://github.com/vibgupta123/NIH-R25-Talks

    - Media Dissemination of Knowledge
      - Magazine articles
        - Written article published in CXOTech magazine on the role of mHealth and Wearables in Disease Diagnosis in Healthcare, 2024, Link: https://cxotechmagazine.com/role-of-mhealth-and-wearables-in-disease-diagnosis-in-healthcare/
        - Article published in The Chronicle of Higher Education, Adapting to AI (June 2024) issue. The published article is public but paid.
      - News reports at (UTMB)
        - Rising STARs Award Launches DIVA-AI Lab at UTMB. https://www.utmb.edu/spph/about-us/home/news/2026/06/10/rising-stars-award-launches-diva-ai-lab

        - Multimodal AI for Precision Medicine at H-COR Works-in-Progress Seminar. https://www.utmb.edu/spph/about-us/news/article/news/2026/03/05/multimodal-ai-precision-medicine-hcor-works-in-progress
        - Meet the Experts: Dr. Vibhuti Gupta on applying AI in healthcare research. https://www.utmb.edu/spph/about-us/news/article/news/2026/02/20/meet-the-experts-dr-vibhuti-gupta-on-ai-in-healthcare s
    - News reports at other Universities (Meharry Medical College)
        - Dean Mhlanaga and Dr. Gupta present at NIH Data sci connect workshop. https://sacsmeharry.org/blog/dean-mhlanga-and-dr-gupta-present-at-2025-nih-datasci-connect-workshop/
        - Drs. Hussain and Gupta share generative AI tips at SACSCOC annual meeting. https://sacsmeharry.org/blog/hussain-and-gupta-share-generative-ai-tips-at-sacscoc-annual-meeting/
        - NSF Expand AI Capacity building pilot network supports Meharry research in trustworthy Ai in medical systems. https://sacsmeharry.org/blog/nsf-expand-ai-capacity-building-pilot-network-supports-meharry-research-in-trustworthy-ai-in-medical-systems/
        - Gupta collaborative research to improve the quality of mHealth datasets for AI/ML applications in HCT research. https://sacsmeharry.org/blog/gupta-collaborative-research-to-improve-the-quality-of-mhealth-datasets-for-ai-ml-applications-in-hct-research/
        - Study finds caregivers of HCT patients likely to adopt digital health technologies. https://sacsmeharry.org/blog/study-finds-caregivers-of-hct-patients-likely-to-adopt-digital-health-technologies/
        - Dr. Gupta helps develop method to improve biometric authentication of mHealth devices. https://sacsmeharry.org/blog/dr-gupta-helps-develop-method-to-improve-biometric-authentication-of-mhealth-devices/

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